Vertically Integrated Industries

Initial takeaways on minerals, metals, manufacturing and chemistry

1. Tailwinds, key takeaways and macro

This theme covers companies that make physical things in their own plants, such as metal parts, magnets and molecules, and that use their own software and AI to make them faster and with fewer skilled people than older competitors. These companies earn money on what they ship. We call them vertically integrated because one company owns the whole job, from taking the order to shipping the product. The takeaways below explain why demand is rising, what makes these businesses hard to build, and where the money ends up.

1. **Some tailwinds behind US production will last and others can reverse, so we should build the thesis on the ones that last.**
   - **Lasting:** China's control of mineral refining and magnets, the shortage of skilled factory workers, and the years it takes a major buyer to approve a new supplier will not change with an election or a budget cycle.
   - **Temporary:** Tariffs, the munitions surge and today's interest rates can all change quickly, because tariffs are set by proclamation and much of the munitions money depends on bills Congress has not passed. The Fed raised rates to 3.75-4.00% in September, and spending on new factory construction was down 21.9% year on year by May 2026 (Federal Reserve, ENR).
   - **Implications:** We should favor companies that still work if tariffs fall and the munitions surge slows. In practice that means companies that own approved capacity buyers will need for a decade, and it means avoiding businesses whose margin exists only because of a tariff.
2. **China's hold on minerals and magnets is the most lasting limit in the theme.**
   - China leads refining in 19 of 20 strategic minerals, with about 70% average share (IEA). Refining is the step that turns ore into pure metal. China also makes about 94% of the world's rare-earth magnets, which go into drones, missiles, motors and robots (Mining.com).
   - The US relies fully on imports for 16 critical minerals (USGS). China has already used export controls on these materials, and its current pause on those controls ends on November 10 and November 27, 2026.
   - Because of this, buyers and the government will pay for non-Chinese supply even when it costs more.
3. **Defense demand and government money are real, but much of the money is conditional and slow to arrive.**
   - The FY27 budget request asks for $49.5B for 10 key munitions, about 8.5 times the average of the past decade, but about 82% of that depends on a reconciliation bill Congress has not passed (CNAS). The Navy is also putting $900M into plants that make submarine parts (Breaking Defense).
   - The FY26 defense bill lets the Pentagon buy munitions over several years and commit to purchases of up to 10 years with advance payments. This is new and helpful, although Congress still has to approve the money each year (FY26 NDAA).
   - The government now acts as a lender and investor. The Pentagon's Office of Strategic Capital (OSC), which lends to companies in key supply chains, has made a $620M loan to one magnet maker and a $500M conditional loan to a rare-earth metals maker, and about $20B is requested for it in FY27 (Mintz). The Department of Energy's loan office has over $289B of lending authority (DOE). These loans take a long time to close and come with conditions.
4. **The labor gap will last, and it is pushing owners of small shops to sell.**
   - US manufacturers need to fill about 3.8M jobs by 2033, and about 1.9M of those jobs are projected to go unfilled (Deloitte).
   - Of the 17,156 machine shops in the US, 83% have fewer than 20 employees (Census). Many owners are near retirement with no successor, which puts a steady supply of certified shops up for sale.

- When skilled people are scarce, a plant that needs fewer of them can grow where its competitors cannot. That is the opening for software and automation.

5. **Spending capital before contracts are signed is the central risk, because a factory has to spend its money before it wins the orders.**
   - Capex means capital spending on buildings, machines and certifications. A major defense contractor, often called a prime, will not approve a new supplier or sign a volume contract until the plant exists and is certified. Approval takes about nine months to two years, based on what two primes told us in expert calls.
   - One precision-parts maker committed about $200M to a single Arizona plant (Plant Services), and its CEO says its capex runs about two years ahead of revenue (claim). Process plants need to run at about 80-85% of capacity to make money (FTI Consulting).
   - This hurts venture returns, because equity spent on a building earns building-like returns and dilutes the founders before the first volume order arrives.
   - The companies that succeed shift who pays for the build. They use government loans and grants, buyers that agree to purchase output before the plant is built, customers that host the factory inside their own plants, bank debt once revenue exists, leased buildings, franchising, and acquisitions of plants that are already certified.
6. **Buyers pay for speed and spare capacity, and few pay for a lower price, so the margin goes to whoever owns the output and holds the approvals.**
   - "Cost is secondary. Cost is driven usually by schedule," a director at a defense prime told us in an expert call, and an engineer there said he would pay 20-40% more to avoid a delay. The premium is highest on prototypes, spare parts and urgent orders, and it fades at volume.
   - Owning the plant pays. One listed parts maker earns a 49% gross margin in its own plants and 31% on work it sends to partner shops (company filings). Approved incumbents also earn well: one engine-parts unit runs a 37.7% EBITDA margin (earnings before interest, taxes, depreciation and amortization). Because aerospace programs can run about 50 years, an approval keeps paying for decades.
   - Selling hardware to factories has been hard. One 3D-printer maker fell from a $2.1B valuation to a $42.5M sale (3D Printing Journal), and an AI-run machining startup in Germany went insolvent in August 2026. Buyers also cap each supplier at about a quarter to a third of their work, so even a winning supplier grows one share at a time.
7. **Today AI helps most with the work around the machine and with processes that run nonstop, but nobody has shown yet that it makes each unit cheaper as volume grows.**
   - The work around the machine includes quoting, planning, writing the instructions a machine follows (CAM programming), inspection and paperwork. One startup claims it cut the time to write assembly instructions from about 10 hours to 15 minutes (claim), and a former employee of one parts maker said its time from order to first part fell about 50% over two years.
   - The machines themselves sit idle much of the time. In a survey of more than 100 plants, cutting machines ran about 34% of the time against a 70% target, and European process plants run at about 74% of capacity against the 80-85% they need (FTI Consulting). Planning and control software can close that gap without a new plant.
   - In a plant that runs nonstop, such as a chemical or metals line, every extra point of yield (good output per unit of input) is close to pure profit. Labor is under 10% of the cost of goods in a new automated plant (claim), so the case for automation rests on speed and yield more than on headcount.
8. **The key risk is that incumbents automate and acquire faster than new companies can grow.**
   - The primes decide which suppliers are approved and how much work each one gets. The largest aerospace parts supplier saw its share of one buyer's machining fall from about 50% to 30%, and it is now automating (expert call).

- Incumbents often buy instead of build. Miners back startups through their venture arms, one miner paid $6.7B for a lithium producer, and a defense contractor's venture arm led a round in an AI materials startup. Drug companies are building their own AI labs, and one has plans to spend up to $1B.
- New companies still have room. The Pentagon relies on more than 200,000 suppliers and has country-of-origin data for fewer than 10% of the lower-tier suppliers on the F-35 (GAO). Most of that supplier base is small, old and short of people, and that is where new companies can win.

2. Interesting categories

This section follows the categories in our primer. For each category we set out the main tailwinds and headwinds, explain whether we should focus there and why, and list the questions we would want answered first.

A. Raw Materials

Stage A: Locating & finding minerals and mines

- **Tailwinds:** Miners need new deposits, and one explorer's CEO estimates that about 650 more mines are needed by 2040 for five battery and magnet minerals (claim). AI explorers with their own rigs and labs say they get drill results back in about 5 days, against 5 months before (claim).
- **Headwinds:** Most finds never earn anything. About 40 deposits are found each year and about a fifth of them become mines (claim), and one explorer puts the industry's success rate at about 0.5% per project (claim). The wait is also long: mines average 15.7 years from discovery to production, and US mines opened in 2020-24 took 19.1 years.
- **Why we should not focus here:** An explorer that does not own the mine does not own the economics. Explorers usually bring in a miner through an option or joint venture, in which the miner funds the drilling to earn 51-70% of the project. A typical deal pays the explorer $1M in cash and $1M in shares over 3-5 years for 60% of the project, leaving the explorer with 40% and a 2% royalty (one explorer's published terms). That royalty, usually a net smelter return (NSR), is 1-3% of metal sales after transport. Early ground sells for little, such as C$200,000 plus a 1% royalty in one 2025 sale, while a 1% royalty on a producing gold mine sold for US$9.9M (company releases). A royalty pays real money only once someone else has built the mine. An explorer that keeps the mine turns into a capital-heavy mining company instead: one AI explorer owns 80% of a Zambian copper project that needs about $2.3-2.5B and targets first copper in the early 2030s (Reuters, Ecofin). In both cases the return is too small, too late or too capital-heavy for a venture fund. Selling software to miners does not fix this either, since one explorer ran about 80 pilots that never turned into recurring revenue.
- **Biggest questions to answer first:** Has any AI explorer sold a proven deposit, and how did the price compare with what it spent to find it? Do option payments cover an explorer's drilling costs before any mine is built?

Example companies: Earth AI, KoBold Metals, Terra AI, GeologicAI, Ideon

Stage B: Mining & refining

- **Tailwinds:** Defense and auto buyers need a non-Chinese source of metal and magnets, and the government is paying to build one: the Pentagon made a $620M loan to one magnet maker and agreed to buy all of another plant's output for 10 years with a price floor. Restarting an old, permitted mine skips much of the permit wait, and one startup says its restarted copper mine runs about half autonomously (claim).
- **Headwinds:** Chinese refineries take about 6 months to commission, against 2 to 4 years in the West. A Western plant takes about 5 years to build and another 3 to 5 to ramp after permits (claim), and it costs hundreds of millions of dollars before it earns any revenue. Without a price floor, a US producer sells at prices set in China, and the largest US rare-earth producer lost $85.9M on $224.4M of revenue in 2025.
- **Why we should focus only on a narrow slice:** We should focus only on refiners and magnet makers where government debt pays for most of the plant and a buyer has signed for the output at a floor price. Those

deals move the capital and price risk off the company, so venture equity pays for the team, the process and the first line. We should pass on miners and refiners without that support, because they carry Chinese price risk on a plant that equity alone cannot fund.

- **Biggest questions to answer first:** How much of the plant is funded by government loans or grants, and what conditions come with that money? Is there a signed purchase agreement with a price floor, and how many years does it cover? What will the unit cost be at full output compared with the Chinese price?

Example companies: Mariana Minerals, BHP, Rio Tinto, Caterpillar, Vulcan Elements, Phoenix Tailings, ReElement, Ucore, Energy Fuels, MP Materials, Hertha, Cyclic, Mass Magnetics, Duranium, Boston Metal, Electra, Ceibo, Nucor, Carpenter Technology

B. Manufacturing

Metal Supply

- **Tailwinds:** Tariffs and buy-American rules help US metal, and a leading US steelmaker credits trade policy for record shipments. Job shops also want faster service than brokers give them, and one online distributor says it quotes in minutes (claim).
- **Headwinds:** The business is crowded with established incumbents, and the largest US distributor is gaining share without AI. Distributors earn a spread on the metal plus cutting fees, and much of their cash sits in inventory. Online ordering with instant quotes already exists, so technology does little to set a new distributor apart.
- **Why we should not focus here:** The margin in metal supply stays with the incumbents, who already have the stock, the buying power and the customers. A new distributor ties up its capital in inventory and then competes on price and availability, where the incumbents are bigger. The tariff tailwind mostly helps the mills and distributors already in place.
- **Biggest questions to answer first:** Does any new distributor earn better margins or turn its inventory faster than the leader at scale? Do faster quotes keep customers coming back, or do they switch on price?

Example companies: Nox Metals, Reliance, Ryerson

Fabrication

- **Tailwinds:** Fabrication means cutting, bending and welding metal into frames and enclosures, and assembling parts like wire harnesses, which are the bundles of wires that carry power and signals through an aircraft or missile. Buyers pay for speed, and one prime's engineer said he would pay 20-40% more to avoid a delay. The leading companies already have revenue: one online fabricator handles about 32,000 orders a month, is growing about 100% a year and says it has been profitable for eight years (claim).
- **Headwinds:** The speed premium fades at volume, and one founder says he beats China on price on only about 20-25% of his work (claim). Customers that grow often buy their own machines and take the work in-house, and buyers cap each supplier at about a quarter to a third of their work.
- **Why we should focus here:** Fabrication plants cost far less than precision plants, so a company can grow on its own revenue. Owning the plant also pays, since one listed parts maker earns a 49% gross margin in its own plants against 31% on work it sends to partner shops (company filings). Low capital needs, existing profits and buyers who pay for speed make this one of the best fits for a venture fund in the theme.
- **Biggest questions to answer first:** What share of revenue comes from repeat customers, and do the largest customers keep buying as they grow? What gross margin does the company earn in its own plants compared with work sent to partners? How much of the work is prototypes and urgent orders, where the speed premium holds?

Example companies: SendCutSend, 1872, Senra Systems, Machina Labs, CircuitHub, Jeevy, Protolabs

Precision & Defense Hardware

- **Tailwinds:** Precision parts are metal or composite parts made to very tight limits, for missiles, jets, satellites and submarines. Munitions and submarine budgets are rising, and the Pentagon can now commit to purchases of up to 10 years. Primes earn 7-12% on cost-plus contracts and outsource much of this work to small, slow shops, many of which have retiring owners.
- **Headwinds:** A new plant has to be built and certified before it wins volume work. AS9100, the aerospace quality standard, typically takes 6-12 months, and approval at each buyer takes about nine months to two years. Much of the munitions money still depends on Congress.
- **Why we should focus here:** An approved supplier keeps the work for programs that can run 50 years, and approved incumbents earn EBITDA margins as high as 37.7%. The factory model decides whether a venture fund can reach that margin. A company that builds its own factory from scratch keeps the whole margin and all the data, but it pays for the plant before approval: one parts maker committed about $200M to a single plant. Buying certified shops from retiring owners is cheaper, and roll-ups buy them at about seven times annual earnings (claim). In a franchised network the franchisees buy the machines and the company supplies software, orders and quality rules; one claims 4 owned and 10 franchised factories (claim). Cells hosted inside a buyer's plant share the cost with the buyer, as in one 7-year deal inside a prime's missile plant. We should focus on the acquired, franchised and hosted models, which keep less of each sale but need far less capital, and back built-from-scratch plants only when contracts or government money cover the build.
- **Biggest questions to answer first:** How long does it take from opening a plant to the first volume contract, and how much cash is spent in between? What share of revenue comes from multi-year contracts rather than one-off purchase orders? Does the company's software raise margins at the shops it acquires?

Example companies: Hadrian, Isembard, Divergent, Freeform, Amca, Atomic Industries, Fabri, Layup, Hybron, Rangeview, Furientis, Helicon, Westmag, Re:Build, Anduril, Precision Castparts, Howmet

C. Specialty Chemicals

Producers (hybrid)

- **Tailwinds:** We limit chemicals to automated labs that make new molecules for drug programs. Drug companies send most of this work to China today, but BIOSECURE, a law passed in December 2025, limits the use of certain Chinese biotech suppliers, and 79% of 124 biotechs surveyed had a China-based contract manufacturer (BIO). The older company in this category has run its automated lab since June 2025, has 152 staff and says it works with eight of the top 20 drug companies (claim, SCI), and a trade publication reported that it has 10 big pharma clients (Citeline).
- **Headwinds:** US chemists cost several times Chinese rates, so these labs must automate to reach offshore prices. Neither lab has disclosed revenue, and the newest one launched in October 2026 with only a small number of design partners and no public customers.
- **Why we should focus here, selectively:** These labs make small batches of new molecules for drug teams at milligram-to-gram scale and do not develop their own drugs. They are paid per compound delivered, or through milestones and royalties if a molecule advances, and the one deal with public terms pays this second way. Each lab costs tens of millions to build, and one is part-funding an £89.9M expansion with about £22M of UK and Scottish grants. We should focus on labs that already have paying drug-company work, because paid compounds prove that customers value the service. A lab that relies on milestones waits years for cash.
- **Biggest questions to answer first:** How much revenue comes from paid compounds today, and how much depends on future milestones and royalties? What does a compound cost compared with an offshore provider? How many drug-company clients have come back for a second program?

Example companies: Chemify, Warp Sciences

D. Advanced Materials

Producers

- **Tailwinds:** China makes about 85% of battery cathode material and over 98% of the cheaper LFP type, so buyers need another source. Data centers used 415 TWh of power in 2024 and are on track for about 945 TWh by 2030, with cooling taking 7% to over 30% of that power, which creates buyers for new cooling materials.
- **Headwinds:** Customers take one to five years to qualify a new material. Owned cathode plants cost hundreds of millions of dollars and compete on Chinese prices, and an established incumbent's battery-materials unit lost money on €436M of revenue.
- **Why we should focus only on a narrow slice:** These companies design the material with AI in their own lab, then make it and sell it by the kilogram or ton under supply deals, so they keep the margin. Some own the plant, and others use a contract maker, which cuts the capital needed. One says its cooling fluid reached the market in about 3 years, against roughly 10 years and $100M for a typical new material (claim). We should focus on companies that use contract makers and already have a named buyer qualifying the material, and pass on companies that need an owned plant to compete on Chinese prices.
- **Biggest questions to answer first:** Which buyer is qualifying the material, and what volume and price follow if it passes? Who will make the material, and at what cost per kilogram? Does the business still work if policy support for non-Chinese supply ends?

Example companies: Orbital Materials, Mitra Chem, Chemix, Albemarle, Umicore, Corning

Materials Discovery Labs

- **Tailwinds:** AI and robotic labs can test far more candidates than people can. One lab made about 1,200 alloys in five or six months (claim), and a university test found that one of them beat a standard aerospace alloy. Funding is available for strong teams, from a $300M seed round to one lab reported at about $7.5B.
- **Headwinds:** None of these labs sells material at volume yet, and finding a material faster does not shorten the years customers take to qualify it. Today the labs earn from paid R&D, licensing, software and government research awards of around $1M. One investor says materials has had "no AlphaFold moment" yet.
- **Why we should focus here, but on a longer horizon:** A lab that wants to sell material at volume usually has to go through a partner that scales it up and sells it, so the lab earns a fee or royalty unless it builds its own production later. Valuations rest on the team, which makes this category more of an experiment today. We should get to know the teams and wait for a material that a buyer has qualified before we treat any lab as a near-term investment.
- **Biggest questions to answer first:** Has a buyer qualified any material from these labs, and at what volume? Who will make the material, and who keeps the margin? How much does each lab earn per partner today, and does that revenue repeat?

Example companies: Radical AI, CuspAI, Lila Sciences, Periodic Labs, Aionics, Mattiq, Dunia Innovations, Entalpic, Discovered Materials, Kebotix, Citrine Informatics, Uncountable, Albert Invent, Microsoft Discovery

E. Enabling Software & Robotics

Enablers

- **Tailwinds:** These companies sell into the lasting problems from Part 1: plants cannot hire enough skilled people, and their machines sit idle much of the time. Cutting machines run about a third of the time against a 70% target, so software that raises use pays back quickly, and the US installed only 38,428 industrial robots in 2025 against 354,226 in China.

- **Headwinds:** Sales to plants are slow, and incumbent automation vendors bundle software with the hardware they have already installed. Robots are not yet reliable enough for most factory work: robot-learning success is 59-66% on short tasks against the 99.9% or more that a factory needs.
- **Why we should focus here:** There is no factory to build, so the money goes into product and sales, which fits a venture fund. Plant-control software adjusts setpoints such as a temperature or flow rate on its own, and one startup says it is paid on the production it adds (claim). Factory software is sold by subscription, and one tool claims it cuts the time to write assembly instructions from about 10 hours to 15 minutes (claim). We should focus on products that are priced on output or save hours a buyer can count, because those win the second and third plant.
- **Biggest questions to answer first:** Is the price tied to output, and what payback do customers actually see? How long does a sale take to go from pilot to a plant-wide contract? Does a second and third plant follow the first?

Example companies: Lumaril, Arda, Mind Robotics, GrayMatter Robotics, CloudNC, AI Robotics, Bright Machines, Rapta, Allus, Faraday, Siemens, Rockwell Automation, Honeywell, AspenTech, Emerson